

BOTOSANI, Romania

WMO: 150200

Lat: 47.7356N

Lon: 26.6456E

Elev: 162

StdP: 99.39

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.6	-13.0	-19.0	0.7	-15.9	-15.7	1.0	-12.0	11.1	1.4	9.4	-0.9	1.4	300	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	32.4	21.0	30.6	20.4	29.0	19.6	22.5	29.2	21.6	28.1	20.8	26.9	2.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.5	15.5	25.2	19.6	14.6	24.2	18.7	13.8	23.2	67.0	29.0	63.7	28.2	60.6	27.0	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.8	6.5	-20.4	35.1	3.9	2.2	-23.2	36.7	-25.5	38.0	-27.8	39.2	-30.6	40.8		
(5)			-20.6	24.2	3.9	1.1	-23.4	25.0	-25.7	25.6	-27.9	26.2	-30.7	26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	-2.2	-0.2	4.5	10.9	16.3	19.8	21.6	21.0	16.0	10.1	4.9	-0.6		
	DBStd	9.51	5.69	5.99	4.93	4.04	3.67	3.15	2.83	3.09	3.70	4.32	4.84	5.49		
	HDD10.0	1431	380	286	178	37	2	0	0	0	2	52	166	329		
	HDD18.3	3264	638	518	428	226	83	21	6	9	87	257	404	587		
	CDD10.0	1513	0	1	8	62	198	293	361	341	183	54	12	1		
	CDD18.3	304	0	0	0	1	20	64	108	92	17	2	0	0		
	CDH23.3	2971	0	0	0	23	234	609	998	894	196	16	0	0		
	CDH26.7	993	0	0	0	4	49	183	359	346	51	2	0	0		
(14) Wind	WSAvg	2.5	2.5	2.7	3.1	2.9	2.6	2.4	2.3	2.1	2.3	2.3	2.6	2.4		
(15) Precipitation	PrecAvg	591	20	23	31	53	67	84	96	64	49	46	31	27		
	PrecMax	805	51	48	71	151	156	172	214	181	152	189	69	75		
	PrecMin	353	5	6	7	8	12	19	11	25	6	7	1	0		
	PrecStd	128	11	12	18	32	40	40	57	40	37	37	17	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	15.4	20.4	26.1	30.4	32.5	34.7	35.1	30.7	25.3	19.8	13.3		
		MCWB	6.9	9.6	11.4	15.4	18.2	21.0	20.8	21.8	19.1	16.7	13.2	8.6		
	2%	DB	8.8	11.9	17.1	23.0	27.9	30.5	32.4	32.6	27.9	22.0	16.3	10.2		
		MCWB	5.1	7.6	10.1	14.2	17.5	20.5	21.1	21.1	18.1	15.3	11.8	7.0		
	5%	DB	6.5	9.2	14.4	20.6	26.0	28.8	30.4	30.4	25.5	19.3	13.8	8.0		
		MCWB	3.9	6.0	8.6	13.0	17.1	19.6	20.7	20.5	17.2	14.0	10.5	5.6		
	10%	DB	4.8	7.1	12.1	18.4	23.9	27.1	28.5	28.2	23.1	16.9	11.8	5.9		
		MCWB	2.9	4.6	7.3	11.7	16.0	19.1	20.0	19.8	16.4	12.6	9.4	4.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	10.0	12.1	16.8	20.1	23.2	23.6	23.8	20.9	17.9	13.9	9.4		
		MCDB	11.0	14.1	19.1	24.5	26.7	29.2	30.0	31.1	28.8	24.0	18.6	12.4		
	2%	WB	5.6	7.9	10.5	15.1	18.7	21.9	22.4	22.5	19.2	16.0	12.3	7.4		
		MCDB	8.0	11.5	16.1	21.9	25.4	28.0	29.0	29.5	25.1	21.1	15.5	10.0		
	5%	WB	4.2	6.4	9.2	13.7	17.7	20.9	21.6	21.4	18.0	14.6	11.0	5.8		
		MCDB	6.1	8.9	13.8	19.1	24.3	26.5	28.1	28.1	23.6	18.4	13.4	7.8		
	10%	WB	3.0	4.8	7.8	12.4	16.7	19.9	20.7	20.4	17.0	13.2	9.6	4.3		
		MCDB	4.6	7.0	11.4	17.2	22.8	25.4	26.9	26.7	22.1	16.5	11.7	5.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	6.8	8.7	10.5	11.5	11.0	11.4	11.7	10.7	9.4	6.7	6.0		
		MCDBR	8.3	10.3	13.2	14.4	15.1	14.0	14.9	15.6	15.4	13.7	10.5	8.9		
	5% WB	MCWBR	5.7	6.6	7.5	7.2	6.5	5.7	5.5	5.8	6.6	7.5	6.4	6.2		
		MCWBR	7.5	9.7	12.4	13.1	13.4	12.1	12.8	13.5	12.9	12.4	9.3	8.3		
(40) Clear-Sky Solar Irradiance	taub		0.319	0.350	0.387	0.447	0.419	0.417	0.446	0.452	0.408	0.373	0.345	0.317		
		taud	2.278	2.229	2.212	2.086	2.241	2.279	2.203	2.179	2.259	2.336	2.352	2.330		
	Ebn at Noon		737	793	821	803	846	849	815	790	791	761	704	695		
		Edh at Noon	77	100	120	150	133	129	137	135	113	90	72	66		
(44) All-Sky Solar Radiation	RadAvg		1.14	1.90	3.12	4.44	5.64	5.94	5.90	5.31	3.71	2.35	1.15	0.87		
	RadStd		0.17	0.28	0.33	0.40	0.46	0.50	0.42	0.39	0.46	0.30	0.10	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (31 neighbors)	+0.68	N/A	N/A	+0.84	N/A	N/A	-146	-206	+103	+34

Nomenclature: See separate page